A Tiny Adventure into the World of
SYNTHETIC BIOLOGY

NAME : _____

CAN WE TURN BANANA INTO ORANGES?





Hi, I'm Frosty! What do you think? Can a banana be changed to become more like an orange?

- ☐ YES
- ☐ NO
- ☐ MAYBE... BUT HOW?!

FIND THE DNA!  *# Misson 01*

Can we actually SEE DNA? Let's find out!

MASH → *BREAK THE CELLS* → *FILTER* →

ADD ALCOHOL → *DNA APPEARS!*



SOAP

Breaks down the cell membranes.



SALT

Helps the DNA molecules come together.



ALCOHOL

DNA does not dissolve well in alcohol, so it forms a visible precipitate.



LOOK CLOSELY! Can you spot the white, stringy material?

OBSERVATION : _____

_____

_____

WHAT IS SYNTHETIC BIOLOGY?

Synthetic biology is like using living things, such as tiny bacteria or yeast, as little factories that we program to make desired stuff like medicines, fuels, or food, the way you'd program a computer.

Meet DNA 

DNA is the molecule that carries instructions for living things. DNA is made from four chemical "letters" — and different combinations of these letters carry different instructions.

A T G C



SCIENTIST CHECKPOINT #1

What do you think would happen if we skipped the alcohol?



- ☐ DNA would be harder to see
- ☐ DNA would turn orange after some time
- ☐ Something would still float on solution
- ☐ We get less DNA



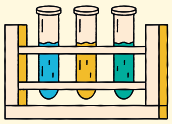
But wait... DNA is only the beginning! Living cells contain lots of different molecules. How can scientists separate them?

LET'S FIND OUT!

SEPARATE THE MYSTERY MIXTURE! # Misson 02



Our sample contains lots of different molecules. But how can we tell them apart?





CHROMATOGRAPHY!

Chromatography is a technique that helps scientists separate different substances in a mixture.
A leaf may look green, but it contains several different pigments. When we use chromatography, these pigments can separate into different coloured bands.

Mixture
↓
Chromatography Paper
↓
Solvent moves upwards
↓
Molecules Separate

SCIENTIST CHECKPOINT #2



WHAT Colours do you observe? write in order

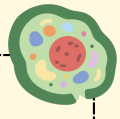
_____
_____
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OBSERVATION:_____
_____
_____
_____

ZOOM INTO THE CELL! # Misson 03



We found DNA. We separated molecules. But where does all of this happen?



INSIDE THE CELL!

Cells are the tiny building blocks of living things. They are too small to see with our eyes, so scientists use microscopes to observe them.
Today, we can take a closer look at cells and discover the tiny structures that make up living things.

PLANT → ***CELL*** → ***DNA***

What We Did???

WE EXTRACTED: Found and isolated actual genetic material.

WE SEPARATED: Discovered how a single mixture holds many different molecules.

WE OBSERVED: Looked deep inside the microscopic world to see real cells.

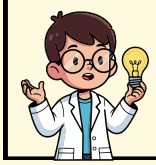
The Big Takeaway:
Living things are built from cells. Cells hold DNA. DNA carries the biological instruction manual for life.

SO, WHAT IS SYNTHETIC BIOLOGY?

Synthetic biology means using our knowledge of DNA and cells to design, engineer, and change biological systems to solve real-world problems.

In simple words:
UNDERSTAND → DESIGN → BUILD → TEST

THINK LIKE A SCIENTIST!



If you could change one thing about a plant or microorganism using biology, what would you change?



There is still so much to discover. Maybe the next big idea will come from YOU!

ONE LAST QUESTION...



So, can we simply turn a Banana into an orange?
Not quite! But scientists can use DNA as biological instructions to understand and change how living systems work.
And that is where the adventure of synthetic biology begins.

OBSERVE → ASK → EXPERIMENT → DISCOVER → DESIGN



Ever wondered how DNA actually builds you?

It all comes down to PROTEINS—the microscopic machines doing all the work in your cells!

Want to build one yourself? Don't just read about it—step into the lab!
SCAN THE QR CODE *to open the simulator. And start building protein by yourself!!*

"It is preferred to view this in desktop view."